Partner-Specific Affective Precision in Social Active Inference

Anonymous Author(s)

Anonymous Institution

Abstract. Social agents must not only infer what partners will do — they must decide how confidently those inferences should drive action. In multi-agent settings this problem is local: one partner may be well-modeled and another barely understood, yet a single global confidence estimate treats both the same. We extend the affective precision mechanism of Hesp et al. [9] to the partner level, giving each relationship its own model-fitness estimate that modulates action confidence independently. We argue that affect, in this setting, is a metacognitive signal: the agent's running estimate of how much each partner model currently warrants decisive action. Using active inference in a repeated multi-partner trust game, we test when partner-local precision helps, when it hurts, and what determines the boundary.
Simulations show that partner-specific affective precision behaves as a social metacognitive signal: it tracks predictive reliability more strongly than realized payoff, improves payoff and reduces policy entropy in graded decision regimes where the policy space is open enough for precision to matter, and shifts engagement toward more predictable partners in partner-choice settings. These effects arise from action selection rather than improved belief quality: the mechanism changes how confidently beliefs drive action, not whether those beliefs are accurate. Partner-local precision is useful when social regularities are stable, neutral when change is gradual, and costly when partners shift faster than first-order beliefs can track. Affect, in this account, is a metacognitive signal that regulates how confidently social inferences are expressed as action—one that is well-calibrated when partner behavior is stable, and reveals its temporal structure when it is not.

Keywords: Active inference · Affect · Social cognition · Multi-agent systems · Trust

1 Introduction

Social cognition involves two separable problems. The first is inference: learning what a partner is likely to do. The second is commitment: deciding how confidently those inferences should drive action. A well-calibrated belief that a partner defects 80% of the time is not the same as knowing how decisively to act on it. The inference machinery and the action-confidence machinery are distinct, and this paper is about the second one.

We argue that affect, in the active inference framework, can be understood as social metacognition: the agent's estimate not of what a partner will do, nor of what they believe, but of how well its own model of that partner is currently performing. This is a distinct computational role. Theory-of-mind inference [7, 13] asks what the partner knows or intends—a second-order model of another mind. Affective precision asks something different: how much should I trust my current model of this partner to guide action right now? The signal is turned inward, toward the agent's own inference quality, rather than outward toward the partner's mental states.

In multi-agent environments this distinction matters practically. A focal agent may have a well-supported model of one partner, a fragile model of another, and an actively changing model of a third. Being precise about a model you have good grounds for, and cautious about one you do not, is not just statistically sensible—it is behaviorally consequential. Over-committing to an uncertain model causes brittle choices; under-committing to a well-validated one wastes hard-won evidence. A predictable adversary and an unpredictable ally may deliver similar realized rewards while presenting very different model-reliability situations; affective precision tracks that difference, not the reward difference.

Active inference is a natural formal setting for this question because it explicitly separates the *content* of beliefs from the *confidence* with which they drive action [4, 5, 12]. Agents infer hidden states by minimizing variational free energy and evaluate candidate policies by their expected free energy [6, 3]. Actions are selected through a precision-weighted policy posterior: the same partner beliefs can produce decisive or tentative behavior depending on the precision parameter γ, which controls how sharply the agent commits to its most-preferred policy.

Hesp et al. [9] provide a formal route from affective valence to inferred model fitness, treating affect as an estimate of how well the agent's generative model is supporting action. Rather than treating γ as a fixed hyperparameter, they model the agent as actively inferring a higher-order belief β over its expected action precision—the subjective fitness of its current generative model. An affective charge signal ϕ drives updates to β: it rises when the model performs better than a prior baseline and falls when it performs worse. This makes affect a derivative of model fitness—a signal that responds to improvements and degradations in predictive performance rather than to reward history—which is what allows it to be positive toward a reliably hostile partner while remaining informative about when to act confidently.

In multi-partner settings this matters particularly. A person can have a reliable model of one partner, a diffuse one for another, and be actively revising a third — and these should not share a single confidence budget. Global precision conflates these situations: it cannot express that you should act decisively with your well-understood colleague and cautiously with a new acquaintance, even when both beliefs are held simultaneously. Confidence, we argue, should be local to the relationship whose model has earned it.

Our extension is to factorize this affective precision signal over social partners. A focal agent interacting with multiple partners maintains a separate belief state and a separate β_k estimate for each partner k, both updated from that partner's behavioral evidence alone. This allows the agent to modulate action confidence differently across its social portfolio: more decisive toward partners its model predicts well, more cautious toward those whose behavior remains uncertain.

We test this in a repeated trust game [2, 10] in which four partners vary in type and disposition across binary and graded decision regimes, partner-choice conditions, and betrayal scenarios. The trust game is a natural testbed because partner types are hidden and must be inferred—so there is a genuine gap between social knowledge and actionable confidence—and because varying the policy space lets us test whether partner-local affective precision operates where the theory says it should.

A key methodological feature is a tracked-only lesion condition that updates β_k fully but fixes γ_k at baseline, severing the link from the precision estimate to action selection while leaving partner-belief updating intact. This design allows us to separate the effect of having a better reliability estimate from the effect of acting on it—a distinction that is easy to conflate in simulation studies without an explicit causal isolation.

We ask three concrete questions. First, does partner-local affective precision track model reliability or realized payoff? Second, does its behavioral effect depend on how open the policy space is, and does the gain trace to action sharpening or to better partner beliefs? Third, when does the mechanism break down—what determines the boundary between adaptive and maladaptive precision modulation?

The value of calibrating action confidence to social model fitness is real and the limits are principled: they tell us something about the timescales on which adaptive social behavior depends.

2 Methods

2.1 Trust game

A focal active-inference agent plays a repeated trust game [2, 10] with four partners over an episode of up to 200 rounds. Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *disposition* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent's payoffs. The structure creates a genuine inference problem: defecting is individually tempting regardless of partner behavior, but a cooperative partner is worth engaging, so the agent must infer the partner's type and disposition to choose well. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. This distributes policy

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	−1
Agent defects	5	1

probability mass across several options, giving partner-local affective precision meaningful room to influence which level is selected. A partner-choice regime adds a prior decision each round: the agent selects which partner to engage before choosing an action, turning approach and avoidance into explicit policy outputs.

2.2 Per-partner generative models

The focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k, rather than a single pooled social model. This keeps evidence about each partner's behavior independent and—critical for the analysis below—allows the confidence with which each model is acted upon to differ across partners. For partner k:

$$s_k = (s_k^{\text{type}},\ s_k^{\text{disp}}) \qquad \text{(hidden: partner type and disposition)} \tag{1}$$

$$o_k = (o_k^{\text{act}},\ o_k^{\text{pay}}) \qquad \text{(observed: partner response and payoff)} \tag{2}$$

$$a \in \mathcal{A} \qquad \text{(cooperate/defect or investment level)} \tag{3}$$

The model $\mathcal{M}_k = \{A_k^{\text{act}},\ A_k^{\text{pay}},\ B_k,\ C_k,\ D_k,\ E_k\}$ encodes likelihoods $(A_k^{\text{act}}, A_k^{\text{pay}})$, transition dynamics (B_k), prior beliefs (D_k), preferred observations (C_k), and policy priors (E_k). Payoff enters as an observation modality rather than an external reward cache, so the agent's preferences are part of the same inference problem as its partner beliefs. State and policy inference use the official `pymdp` discrete active-inference implementation [8, 3].

2.3 Partner-local affective precision

Hesp et al. [9] treat the policy precision parameter γ as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over its expected action precision, representing subjective model fitness—how well the generative model is supporting confident action. An affective charge signal ϕ drives updates to β: it rises when the model performs better than a prior baseline and falls when it performs worse, making it a derivative of model fitness that responds to changes in predictive performance rather than to absolute reward level.

We extend this to the per-partner case. After interacting with partner k, the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P\Big(o_k^{\text{act}} = \hat{o}_k^{\text{act}}\ \Big|\ q(s_k^{\text{type}}, s_k^{\text{disp}})\Big)\,. \tag{4}$$

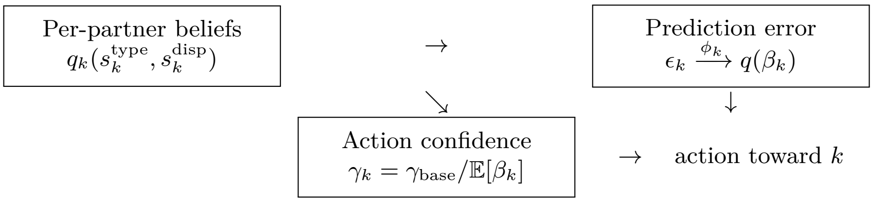

Fig. 1: Partner-local affective precision. Each partner has an independent belief state and precision estimate. Prediction error on partner action updates β_k, which sets how sharply the agent commits to policies involving that partner.

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-disposition model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. We set the neutral fitness baseline at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge.

Affective charge is:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big)\,, \tag{5}$$

where σ_0^2 is a prior fitness baseline and α is a gain parameter. Positive ϕ_k means partner k's model predicted better than the baseline; negative ϕ_k means it predicted worse. A categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$ is updated each round by ϕ_k. Higher β_k represents lower expected model fitness—the agent has accumulated evidence that its model of this partner is unreliable. The partner-specific policy precision is:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{6}$$

Note the direction: β_k represents precision *volatility*, not precision itself. Higher β_k means the model has been less reliable, producing a broader, less committed policy distribution. This runs opposite to the intuition that higher β equals more confidence — it is the inverse that drives action sharpening. A well-validated partner model drives $\mathbb{E}[\beta_k]$ toward low values, raising γ_k and sharpening policy selection for that partner. An unreliable partner model keeps $\mathbb{E}[\beta_k]$ high, lowering γ_k and leaving policy selection diffuse. The mechanism thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

Tracked-only condition. To establish that any behavioral effect of the signal runs specifically through action confidence—rather than through indirect effects on belief quality—we include a tracked-only variant. This variant retains the β_k update in full but fixes $\gamma_k = \gamma_{\text{base}}$ for all partners at all times, severing the link from the precision estimate to the softmax temperature. The β_k posterior evolves normally and is available for analysis, but it never changes how policies are selected. If tracked-only behavior matches the no-affect baseline while full precision modulation differs from both, the behavioral effect runs through the γ_k pathway. If tracked-only matches full precision modulation, the effect is mediated by belief quality, not action sharpening.

3 Results

3.1 Partner-local affective precision tracks partner predictability, not partner value

The central question is whether the β_k signal actually reflects predictive reliability or whether it is a proxy for experienced reward. A predictable exploiter is low value but easy to model; a valuable but erratic partner is worth engaging but hard to predict. If affective precision tracks value, these two situations should produce similar β_k levels. If it tracks model fitness, they should diverge.

We evaluate this in the binary game across 30 seeds and 200 rounds per seed. The γ_k-derived signal correlates more strongly with partner predictability than with payoff: $|\mathrm{corr}(\mathrm{precision}, \mathrm{surprise})| = 0.701$ versus $|\mathrm{corr}(\mathrm{precision}, \mathrm{payoff})| = 0.419$ across partner-seed units. The seed-level gap between these two correlations is $+0.096$, bootstrap CI $[0.027,\ 0.164]$. Partner-local affective precision carries model-reliability information, not reward history.

The absence of a payoff benefit is itself informative: it prevents the precision signal from being explained away as a proxy for reward. In the binary regime the expected free energy gap between cooperating and defecting is large enough that the policy posterior is already near-deterministic; sharpening or softening a near-argmax changes almost nothing. Affective precision is informative about reliability but has nowhere to act. Section 3.2 shows what happens when it does.

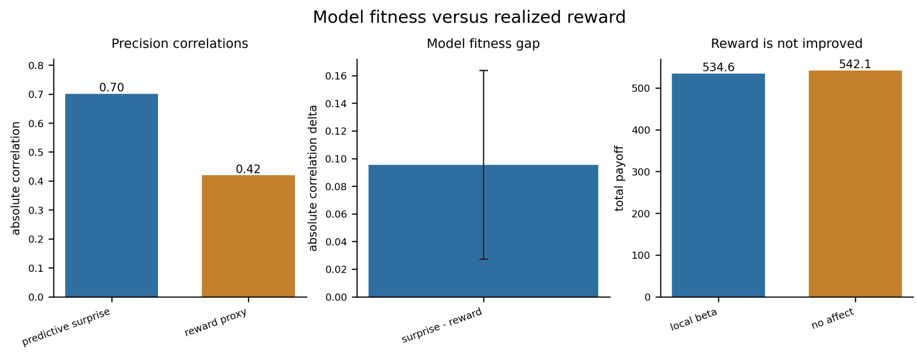

Fig. 2: Partner-local β_k tracks partner predictability more strongly than realized reward. In the binary game the mechanism has no payoff advantage, consistent with a near-saturated policy space where precision cannot influence action.

3.2 Behavioral gains are regime-dependent and trace to action sharpening, not belief quality

Whether affective precision can improve behavior depends on how much room the policy space gives it to act. We compare five seeds per variant, 200 rounds, across both regimes. In the binary game: no payoff advantage, no entropy difference (Section 3.1). In the graded investment game: precision modulation reaches total payoff 1884.6 versus 1859.4 for the no-affect baseline (+25.2), with mean policy entropy 7.94 versus 8.80 (−0.86). The improvement emerges precisely in the regime where policy entropy is high enough for action confidence to matter.

The tracked-only lesion isolates where the effect lives. It matches the no-affect baseline in payoff and entropy — not the full-precision agent — despite having slightly *better* partner-belief accuracy (joint: 0.339 vs. 0.336; stance: 0.855 vs. 0.816). The benefit is not epistemic: the agent is not forming better partner models. It is metacognitive: the agent deploys the same models with better-calibrated confidence. The gain traces specifically to the $\beta_k \to \gamma_k$ pathway.

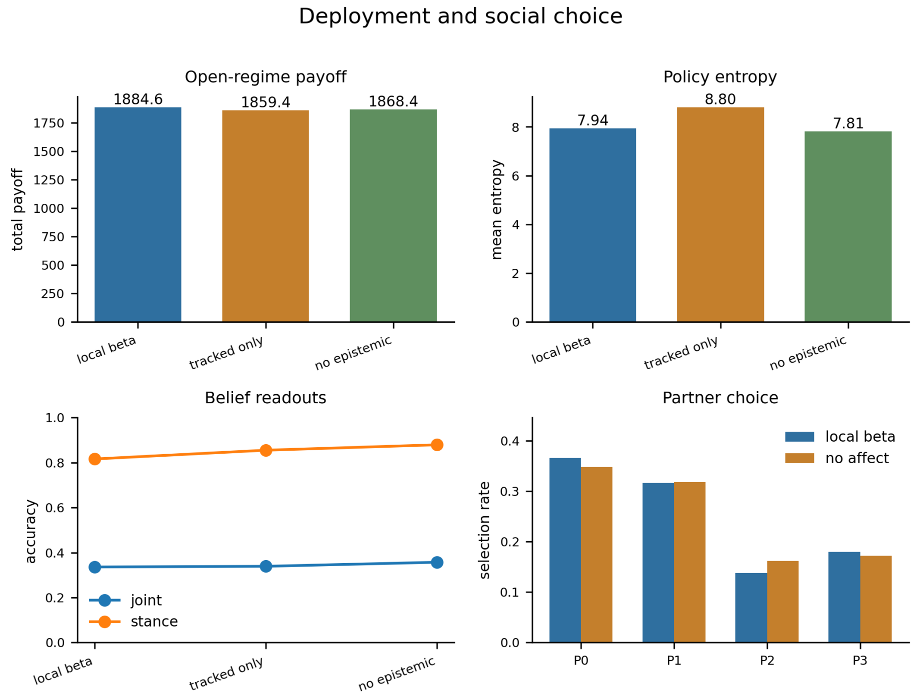

Fig. 3: Behavioral effects are regime-dependent and traceable to action sharpening. Precision modulation improves payoff and reduces policy entropy in the decision-open graded regime only. The tracked-only lesion matches the no-affect baseline despite maintaining comparable partner beliefs, identifying action confidence—not belief quality—as the locus of the effect.

3.3 Partner selection reorganizes around reliability

In partner-choice regimes the agent's action-confidence estimates shape not just how it behaves toward a chosen partner, but which partners it chooses to engage in the first place. Aggregate payoff is a blunt measure here; subtle shifts in selection patterns can reflect meaningful reorientation before payoff has had time to accumulate.

The five-seed pair check shows this deployment effect without a reward advantage. Across 200 rounds per seed, aggregate payoff is nearly flat: 393.6 with precision modulation versus 393.2 without it. Policy entropy is lower under precision modulation (3.99 vs. 4.83), and selection shifts in the expected direction: the most predictable partner draws more engagement ($0.348 \to 0.366$), while a less predictable partner draws less ($0.162 \to 0.138$).

The mechanism is reorganizing social engagement around partner-model reliability. It concentrates interaction where the agent can act confidently, at a scale visible in selection behavior before it registers in payoff.

3.4 Abrupt betrayal is the mechanism's volatility limit

All conditions above involve stationary partners. Betrayal scenarios test what happens when a partner's disposition changes mid-episode: the agent enters the post-switch period with a model that has been accumulating positive affective charge under the old behavioral regime.

The problem is precisely that affective precision has been doing its job. The agent has built up action confidence toward the betraying partner based on a model that no longer applies. When the partner changes, β_k still reflects the pre-switch reliability record while the lower-level partner-state beliefs—which need repeated evidence to reorganize—are only beginning to update. Metacognitive confidence sharpens action around a stale first-order model.

Across 30 seeds and 120 rounds this is what we observe. Precision modulation yields lower whole-run payoff than the no-affect and tracked-only baselines: 1136.1 versus 1172.1, mean difference -36.1, bootstrap CI $[-63.7, -10.9]$, $p = 0.017$. Policy entropy is lower under precision modulation (8.38 vs. 8.74), confirming the channel is active and influencing action. Post-switch, the agent returns to the switched partner less often (4.4 vs. 6.1 reencounters; selection rate 0.049 vs. 0.067) but with no calibration improvement: payoff per reencounter 8.76 vs. 8.91, wrong-type rate 0.237 vs. 0.165. The agent acted decisively; its inference did not justify that decisiveness.

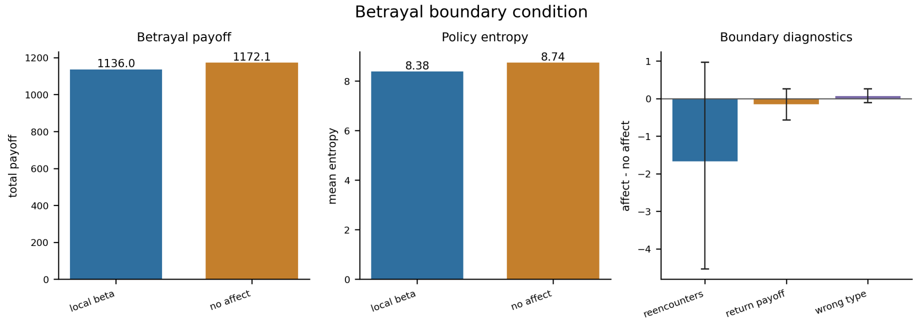

Fig. 4: Abrupt betrayal is the volatility limit. Precision modulation reduces whole-run payoff, lowers policy entropy, and changes reallocation without improving calibration. The agent acts more confidently on a partner model that has become stale.

3.5 The limit is the speed of change, not the amount of precision

The betrayal failure might suggest the problem is too much precision gain—a β_k update that is too fast. The sensitivity analysis rules this out. A combined-caution variant that raises β globally to reduce precision gain produces worse payoff, not better: 1115.0 versus 1172.1 for no-affect ($p = 0.003$, entropy 9.27). Reducing precision uniformly removes the useful action confidence in the stable portions of the episode without addressing the timing problem in the betrayal portion.

Gradual betrayal confirms that the limit is the speed of change. When the partner's disposition shifts slowly, the β_k posterior tracks the changing prediction record before a large confidence surplus can accumulate. Default precision modulation is neutral under gradual change: payoff 1147.6 versus 1148.9 for no-affect, $p = 0.906$. The fact that blanket caution hurts in *both* regimes while local precision is only bad in the abrupt case is a strong interaction: it reveals that the mechanism's limitation is not the magnitude of its response but the temporal alignment between social change and belief recalibration. Combined caution again hurts (1118.3, $p = 0.005$). Partner-local affective precision is adaptive when partner behavior is stable and neutral when it changes slowly; it becomes maladaptive specifically when behavioral change is faster than partner beliefs can track.

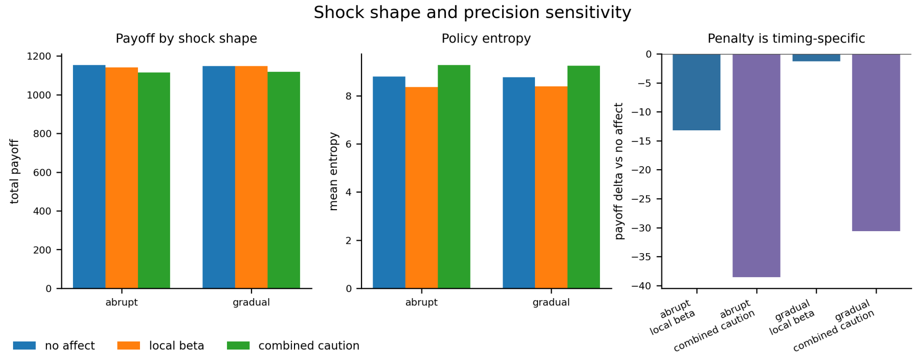

Fig. 5: Shock-shape sensitivity. Default precision modulation is neutral under gradual betrayal and harmful only under abrupt betrayal. Generic caution reduces payoff in both conditions. The mechanism's limit is the timescale of behavioral change, not the magnitude of precision.

3.6 Precision-dynamics phenotypes

As a compact parameter check we verify that β_k dynamics respond to parameterization in an interpretable way. In the open graded regime a low-gain variant (small α) produces narrow β_k movement (mean range 0.149 vs. 0.950 for default); high-gain and cautious-prior variants produce wider movement (1.318 and 1.435). The same ordering holds in betrayal (0.132, 1.114, 1.091). Payoff does not rank variants monotonically by precision-dynamics range, consistent with the regime-dependence result: the behavioral consequence of affective precision depends on the task context, not just on how much the signal moves.

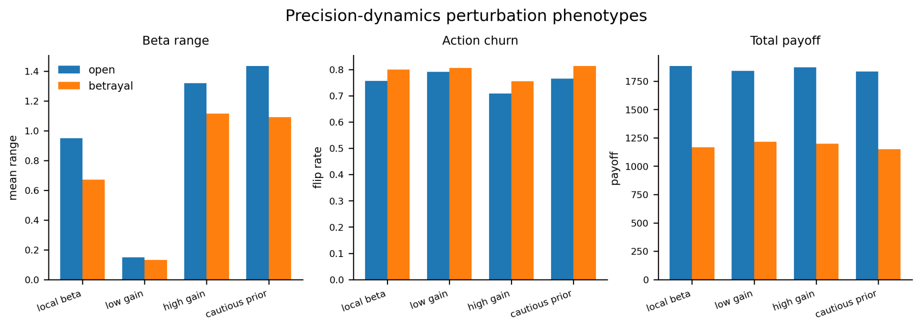

Fig. 6: Precision-dynamics phenotypes. Low-gain, high-gain, and cautious-prior variants separate clearly in β_k movement range. Behavioral consequences depend on the task regime rather than on the magnitude of precision dynamics alone.

4 Discussion

Affect as social metacognition. The results support interpreting partner-local affective precision as a form of social metacognition: the agent's estimate not of what a partner will do, but of how well its current model of that partner justifies confident action. This is a different signal from both reward and belief accuracy. β_k tracks partner predictability, not partner value—a reliably hostile partner and a reliably cooperative one can both earn high β_k; what they share is predictability, not valence.

This distinguishes the mechanism from theory-of-mind inference. ToM approaches model the agent as maintaining representations of what partners believe, know, or intend—a second-order model of another agent's mental states [7, 13]. Affective precision does not do this. It tracks the focal agent's confidence in its own model of a partner's *behavior*, not representations of that partner's mental states. The signal is metacognitive rather than mentalistic: it is about how well the agent's own social inference is currently performing, not about what the partner is thinking.

The two mechanisms can complement each other. An agent with sophisticated ToM inference could still be poorly calibrated about when to act confidently on those inferences—particularly if the partner is reliable in some contexts and unreliable in others. Partner-local affective precision addresses exactly this gap: it allows action confidence to track the recent track record of each partner model, independently of the model's representational content. A natural extension is to combine ToM-style belief inference [14] with partner-local affective precision, so that the agent both models partner intentions and maintains a running estimate of how much those models currently justify decisive action.

The dissociation result supports this. The tracked-only lesion confirms that the behavioral effect is not mediated by better beliefs: remove the $\beta_k \to \gamma_k$ link and the gain disappears, even though partner-belief accuracy is preserved or slightly improved. The agent's social metacognition is doing real work—it is not reducible to better inference or better outcomes. The benefit is not epistemic—the agent is not forming more accurate partner models. It is metacognitive: the same models, deployed with better-calibrated confidence, produce better outcomes. This pattern holds where the policy space is open enough for precision to matter and vanishes where the posterior is already near-deterministic.

The betrayal result reveals the mechanism's temporal structure. Under abrupt betrayal the mechanism does exactly what the metacognitive account predicts—and that is precisely why it fails. Accumulated β_k evidence justifies high action confidence toward the now-changed partner; the agent acts decisively on a model that has become stale. The Hesp et al. architecture has two timescales: affective charge ϕ updates quickly from single-interaction prediction error, while the higher-order state β integrates across many rounds. In stable environments this is fine — β accumulates evidence and converges. Under abrupt change there is a window in which ϕ_k has already registered surprise but β_k still reflects pre-switch confidence. Action sharpens around a model that is no longer valid. This is not a bug in the mechanism so much as an exposure of its temporal assumption: it works when social regularities persist long enough for confidence and accuracy to stay aligned.

The gradual betrayal result confirms this interpretation directly: when behavioral change is slow enough that partner beliefs can track the shift before large confidence surplus accumulates, default precision modulation is neutral. The problem is not how much the signal responds, but how fast the social world changes relative to how fast beliefs reorganize.

The implication is that an effective partner-local precision mechanism in volatile environments would need a way to modulate the weight it places on recent charge signals based on how much the partner's behavior appears to be changing. This is structurally similar to hierarchical volatility tracking in belief-updating models [11, 1], but applied to action confidence rather than to learning rates. Whether these two levels of volatility sensitivity—for beliefs and for action precision—operate independently or are coupled in human social cognition is an open empirical question that this work motivates.

Limitations. Three design features bound the current claims. First, β_k is an external auxiliary tracker; in the full Hesp et al. hierarchy it is a proper inferred hidden state, allowing the agent to have prior expectations about its own future affective states and to plan around anticipated changes in partner reliability. Making β_k a variational variable would enable richer prospective regulation of action confidence. Second, the current implementation keeps $q(\beta_k)$ categorical rather than continuous. This preserves the inverse-beta precision convention and keeps the update identifiable in small simulations, but it discretizes the precision dynamics. Third, partners are scripted rather than themselves being active-inference agents; the mechanism is tested in isolation from the reciprocal dynamics that characterize real social interaction. The global-β ablation—whether partner-local precision is necessary, or whether any shared precision estimate would produce comparable stable-regime gains—is therefore a core robustness check for the final rebaseline.

Future directions. The global-β ablation is the most important near-term robustness check. The predicted outcome is that a shared precision estimate would either over-suppress action confidence toward reliably predictable partners or under-suppress it toward uncertain ones, depending on the mix. A second priority is a controlled shock-gradient study that varies betrayal abruptness parametrically, targeting the timescale boundary at which the mechanism transitions from neutral to harmful and testing the timing-mismatch account directly.

More broadly, the framework invites extension toward richer social environments. The trust game has no communication, reputation signaling, or coalition structure; all of these would introduce new sources of reliability evidence that the affective signal could in principle track. Testing with human participants would allow partner-local β dynamics to serve as a computational model of individual differences in social confidence calibration—with faster, higher-gain channels predicted to show stronger post-cue reallocation but greater sensitivity to abrupt social change. Finally, combining this mechanism with ToM-style belief inference [13] would allow future work to study whether metacognitive action-confidence signals and mentalistic partner-state models interact during social decision-making, and how they trade off under resource or time constraints.

5 Conclusion

Predicting what a partner will do and deciding how confidently to act on that prediction are separable problems. β_k is the agent's running estimate of how much epistemic warrant each partner model currently holds for guiding action—not a measure of partner value, not an improvement to inference quality, but a second-order signal about when and how decisively first-order social beliefs should be acted upon. In repeated trust-game simulations, partner-local affective precision tracks partner predictability rather than partner value, improves performance through action sharpening in decision-open regimes, and reorganizes partner engagement around reliability before aggregate payoff separates. When partners change abruptly, accumulated metacognitive confidence sharpens action around a model that has not yet caught up. Together, these results support a metacognitive account of social affect: a signal that tracks inferential warrant rather than social value, and whose adaptive profile—useful under stability, neutral under gradual change, costly under abrupt change—reflects the temporal conditions under which social confidence can be well-founded.

References

1. Behrens, T.E.J., Hunt, L.T., Woolrich, M.W., Rushworth, M.F.S.: Associative learning of social value. Nature **456**, 245–249 (2008). https://doi.org/10.1038/nature07538
2. Berg, J., Dickhaut, J., McCabe, K.: Trust, reciprocity, and social history. Games and Economic Behavior **10**(1), 122–142 (1995). https://doi.org/10.1006/game.1995.1027
3. Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., Friston, K.J.: Active inference on discrete state-spaces: A synthesis. Journal of Mathematical Psychology **99**, 102447 (2020). https://doi.org/10.1016/j.jmp.2020.102447
4. Friston, K.: The free-energy principle: a unified brain theory? Nature Reviews Neuroscience **11**, 127–138 (2010). https://doi.org/10.1038/nrn2787
5. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo, G.: Active inference: A process theory. Neural Computation **29**(1), 1–49 (2017). https://doi.org/10.1162/NECO_a_00912
6. Friston, K., Rigoli, F., Ognibene, D., Mathys, C., Fitzgerald, T., Pezzulo, G.: Active inference and epistemic value. Cognitive Neuroscience **6**(4), 187–214 (2015). https://doi.org/10.1080/17588928.2015.1020053
7. Frith, C.D., Frith, U.: Mechanisms of social cognition. Annual Review of Psychology **63**, 287–313 (2012). https://doi.org/10.1146/annurev-psych-120710-100449

8. Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I.D., Tschantz, A.: pymdp: A python library for active inference in discrete state spaces. Journal of Open Source Software **7**(73), 4098 (2022). https://doi.org/10.21105/joss.04098
9. Hesp, C., Smith, R., Parr, T., Allen, M., Friston, K.J., Ramstead, M.J.D.: Deeply felt affect: The emergence of valence in deep active inference. Neural Computation **33**(2), 398–446 (2021). https://doi.org/10.1162/neco_a_01341
10. King-Casas, B., Tomlin, D., Anen, C., Camerer, C.F., Quartz, S.R., Montague, P.R.: Getting to know you: reputation and trust in a two-person economic exchange. Science **308**(5718), 78–83 (2005). https://doi.org/10.1126/science.1108062
11. Mathys, C., Daunizeau, J., Friston, K.J., Stephan, K.E.: A Bayesian foundation for individual learning under uncertainty. Frontiers in Human Neuroscience **5**, 39 (2011). https://doi.org/10.3389/fnhum.2011.00039
12. Parr, T., Pezzulo, G., Friston, K.J.: Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. MIT Press (2022)
13. Pitliya, R.J., Çatal, O., Van de Maele, T., Pezzato, C., Verbelen, T.: Theory of mind using active inference: A framework for multi-agent cooperation (2025). https://doi.org/10.48550/arXiv.2508.00401
14. Ruiz-Serra, J., Sweeney, P., Harré, M.S.: Factorised active inference for strategic multi-agent interactions. In: Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems. pp. 1793–1802. International Foundation for Autonomous Agents and Multiagent Systems (2025). https://doi.org/10.5555/3709347.3743815